Alex Gates — inform status outline (next meeting)

Date: 2026-07-16
From: Charles Levine (COMPASS-assisted skeleton — **you** add detail iteratively)
Purpose: Inform only — map the full path in minimal bullets. **Not** the decision questionnaire.
Questions / tabled items: `20260611_Alex_Gates_Talking_Points.md`
Deep plan (if he asks): `03_Where_we_are_now.md` · `04_Project_story_plain_English.md`

1. One sentence

Cross-domain paper: **inverted-U on leave-one-out peer-pool quality** (Army → basketball → tenure preliminary) → **minimal generative mechanism** (honest claims) → **two predictions** → **Wang-style manuscript** — target **Summer–Fall 2026**.

2. Where we are (snapshot)

Leg	Status	One line
Army	✅ Anchor	CIF + Cox stack mature; AWS is canonical run environment
Basketball	✅ Replication	Inverted-U on `poolq_loo`; D10 export bundle on disk
Tenure	⚠️ Preliminary	Stage 9 binned plot OK for draft; 796 persons / 52 depts inference-ready
Generative model	⚠️ Path II	Talent-only fails; congestion-in-score POC on **pool mean** (axis mismatch stated honestly)
Manuscript	🔲 Next	Staging prose written; **inking Word** is current focus

3. The path (whole arc — briefing outline)

How to read: Top-to-bottom = time order. **Done** = already in hand. **Now** = current focus. **Parallel** = runs alongside draft; does not block first Word pass. **Before submit** = needs to land before we send the paper out. **Later** = explicitly parked unless you or Alex elevate.

Phase A — Empirical pattern across three settings (DONE)

What we proved descriptively: In each domain, advancement (promotion, draft, tenure) rises with peer-pool quality through the middle range, then dips in the most elite pools — an **inverted-U**, not "better peers always help."

Setting	What we measured	Status
Army	Leave-one-out senior-rater pool quality vs promotion (competing risks with attrition)	**Established** — strongest anchor
Basketball	Leave-one-out teammate pool quality (`poolq_loo`) vs NBA draft	**Established** — clean replication
Tenure	Leave-one-out department peer quality vs tenure event	**Preliminary** — binned plot only; smaller inference sample (796 people / 52 departments)

One line for Alex: The **cross-domain stylized fact** is real in Army and basketball; tenure is an honest third panel, not a fourth anchor yet.

Phase B — Close the minimal model (NOW)

Goal: Freeze enough mechanism language that the manuscript can describe *what the model is*, *what the data columns mean*, and *what we are **not** claiming* — without waiting for a perfect simulation.

Step	Plain English	Status
B1. Score equation	Advancement score = ability minus congestion penalty (`S = A − λ·congestion`). This is the **constraint-leg** proof-of-concept Alex asked for — who gets selected when pools are crowded.	**Locked** (Tier 1)
B2. Talent-only null	Simulation with ability only (no congestion) **fails** to reproduce the inverted-U — congestion term is necessary for the generative story.	**Done** (`538D` CELL 10)
B3. Congestion proof-of-concept	Simulation **with** congestion bends curves on **whole-roster pool average** — not yet point-for-point on the same axis as empirical `poolq_loo`.	**Done — axis mismatch stated honestly**
B4. Axis table	Written table: each model quantity → data column → which axis the plot uses → allowed claim for version 1. Stops prose from over-claiming generative replication.	**On disk** (D10 export bundle)
B5. Claim guardrails	Master list: every manuscript claim tagged Supported / Preliminary / Unsupported / Defer (`#07`).	**Written** — VECTOR and Charles check prose against it

One line for Alex: Minimal model = **honest Path II** — generative POC on pool-mean axis + explicit limitations; **not** bin-for-bin match to empirical leave-one-out quality before draft.

Phase C — Draft the manuscript (NOW)

Structure (Wang-style): short opening frame (§0) → empirical phenomenon (§1) → theory (§2–§3) → minimal mechanism (§5) → predictions (§7). **Not** a front-loaded lit review; fuller literature map in Discussion (`#12` §5.2).

Manuscript section (Word outline)	What goes there	Ink status
§0 Opening framing	Problem, gap, contribution preview, roadmap (½–1 page)	Staging §1 ready — ink on **polish pass** after §5→§1→§4
§1 Empirical observation	Three-setting triad + figures (Army CIF, basketball ventile, tenure preliminary)	Staging prose ready; Army figures pending AWS canon
§2–3 Theory	Why finite distinction / local comparison pools matter; **L_net = B – D** framing	Staging + nesting note ready
§4 Tenure setting	Preliminary third leg + limitations	Staging ready; Cell 9 plot OK for draft
§5 Minimal generative model	Talent-only fails vs congestion-in-score; axis table; limitation paragraph	Staging + D10 figures ready
§7 Predictions	Two primary prediction **directions** (see below)	Staging ready; one basketball figure on disk

Ink order (your current plan): write **§5** (generative) first → then **§1** (empirical triad) → then **§4** (tenure) → then **§0** (opening frame, ½–1 page polish) — so mechanism language is settled before empirical claims reference it.

Lit review (v1): §0 + §2–§3 + Discussion — **not** a separate chapter before §1.

Two primary predictions (locked for version 1):

1. **Near-threshold heterogeneity** — the elite-pool dip should hit hardest for **borderline** performers (good but not stars), not the very best or the clearly below-average. One exploratory basketball figure exists; not a full tri-domain proof yet.
2. **Peak shift with global capacity (Λ)** — when an organization has more scarce advancement slots (bigger boards, more picks, more tenure lines), the **top** of the inverted-U should move. **Conceptual / prose hook** for now — not a finished cross-domain figure.

One line for Alex: Draft is unblocked — empirical triad is the spine; generative is a disciplined supplement, not a gate.

Phase D — Parallel work while drafting (does not block Word)

Workstream	What it is	When it matters
Tenure Cox models (`540` Cell 12)	Inferential hazard models beyond the descriptive binned plot — hazard-ratio tables for tenure curvature	**Before submission**, not before first draft. Cell 9 plot is legitimate for draft prose.
Army figure canon	You name canonical cell, run profile, and filenames after **AWS sync** (Cursor repo is a digital twin)	Before Alex-facing Army figure is final
Estimand sentences	Three to five approved sentences: Army **descriptive** binned cumulative-incidence bars (Cell 11) vs **inferential** Cox models (Cell 12) — not the same object	Alex sign-off before final Army methods wording
Manuscript inking	Paste staging prose into `Manuscript_working_outline_v1.docx` per ink map	**Now**

Phase E — Advisor read and submission (TARGET: Summer–Fall 2026)

1. Charles completes first full Word draft (empirical + generative + predictions + limitations).
2. Alex read — revise **claim language** against guardrails, not reopen Path II unless he asks.
3. Close soft gates: tenure Cox table, Army estimand wording, pool-size audit (pre-publication).
4. Submit interdisciplinary science-of-science target (venue TBD).

Opportunity-cost rule: If a task does not improve draft quality, claim honesty, or submission readiness → it stays in the parked queue.

Phase F — Parked queue (LATER — mention only if he asks)

Item	Plain English
525 / UIC deep dives	Extra Army work on prestigious units and senior-rater consistency — mechanism meat if the paper needs it
Network extensions	Talent center-of-gravity, talent paradox, exposure networks — dissertation / manuscript §8 future work
Full leave-one-out generative match	Make simulation reproduce empirical `poolq_loo` curve point-for-point — north-star, parallel to draft
Separate B and D estimation (PD12 P1)	Identify benefit and congestion legs parametrically in all three domains — beyond version 1
Fourth domain (PD12 P4)	Add another empirical setting to stress-test generality

Elevate any parked item only if **Charles or Alex** says so.

4. What to tell him (talk track — expand as needed)

- **Bottleneck shifted:** discovery → **convergence** (model language + manuscript).
- **Paper-first:** empirical triad is the contribution; generative is **minimal POC**, not full LOO replication.
- **Tenure:** third leg is **preliminary** — honest limitations, not equal maturity to Army/MBB.
- **Not blocked on:** generative bin-for-bin LOO match, Fine–Gray, 525/UIC, network §, tenure Cox for *draft*.
- **Phase A done (June):** Tier 1 locks filed; D10 bundle; faculty inference export; staging prose.

5. Parked (mention only if he asks — not seeking decisions today)

Item	One line
525 / UIC	Prestige-org Army depth — revisit if manuscript needs more mechanism meat
Networks	Talent COG, talent paradox — dissertation / later §
LOO generative match	North-star parallel; Path II accepts axis mismatch for v1
Pool-size audit	Pre-publication, not pre-draft

6. If he wants more detail — point, don't recite

Topic	Doc
Full status + dependencies	`#03`
Plain-English story	`#04`
Equations / variables	`#21`
What we can claim	`#07`
Alex score vs. **L_net** (say aloud)	**Appendix** (this doc)
TB-stratify, estimand, Fine–Gray	Talking points doc

7. Meeting notes (Charles — add after)

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Appendix — Alex score vs. L_net = B − D (say aloud)

Purpose: Practice language for "two equations, one project — not the same object."
Canonical nesting: `5-Manuscript/05_Model_Nesting_Note_v1.md` · formula decoder `21_Formulas_and_variables.md` §1–§2.

A.1 The two equations (say them first)

Your framing (net environment):

```
L_net = B(·) - D(·)
```

Alex's framing (selection score):

```
S_i = A_i - λ L_C,LOO,i
```

Same project. **Not the same object.**

A.2 One sentence each

Equation	Say this
****L_net****	"How does the **local peer environment** net out for advancement — upside from strong peers **minus** downside from crowding them out?"
****S_i****	"How does the **selector rank** people — own ability **minus** a penalty for viable-peer congestion?"

A.3 The significant differences (what to stress aloud)

1. Reduced form vs. selection rule

	****L_net = B − D****	****S_i = A_i − λ·L_C****
Asks	Why does peer context help *and* hurt?	Who gets picked when pools are crowded?
Level	Environment → advancement propensity	Individual → ranking score → selection
Role in paper	Conceptual spine + empirical story	Minimal **generative** proof-of-concept

Say: "Mine is the **why** — benefit and constraint in the environment. Alex's is the **who gets selected** rule that puts congestion into the score."

2. Two legs vs. one leg in the equation

****L_net**** has **both**:

- **B** — visibility, norms, minutes, development upside from elite peers
- **D** — congestion, substitutability, finite slots

Alex's score has **only the D-leg in the formula** (plus own ability):

- Congestion **L_C** is subtracted from **A_i**
- **B is not a term in** S_i**** — no "minutes bonus" or "development benefit" in the score equation

Say: "Alex's equation is **not B − D**. It's **ability minus congestion** — the constraint leg **entering selection**. The benefit leg lives in my broader frame and in **separate empirical columns** (`poolq_loo` vs `crowding_smooth`), not inside his score."

Nesting line: Alex score = **D entering the selection rule**; not the full **L_net** and not a second model.

3. Own ability: outside vs. inside

- In ****L_net****, own ability ****A_i**** is **outside** the decomposition — environment is about **peers**, not you.
- In ****S_i****, own ability is **inside** the score — ranking is explicitly **you vs. congestion**.

Say: "**L_net** is about **the pool around you**. **S_i** is about **how evaluators score you** given that pool."

4. What you measure empirically vs. what the sim uses

Construct	Empirical world	Alex generative world
Pool "quality"	`poolq_loo` (LOO teammate quality) — **mixes B and D**	Often **pool mean** on roster; LOO quality is a different knob
Congestion	`crowding_smooth` — **D proxy**	$L_{C,\mathrm{LOO}}$ — viable-peer density, LOO
Outcome curve	Inverted-U on **LOO quality**	Inverted-U on **pool mean** when **λ > 0**

Say: "Empirically we plot advancement vs **leave-one-out pool quality**. The generative sim shows congestion **in the score** can bend curves on **whole-roster pool mean**. Same story at the mechanism level, **different conditioning object** — and we say that honestly."

5. What each equation is *for* in v1

- ****L_net: Explains the **inverted-U** (both legs), guides **Rung 2.5** (quality vs. congestion columns), frames **predictions**. **Not separately estimated** as **B(Q)−D(Q)** in v1.
- ****S_i: Shows **talent-only fails** (**λ = 0**) and **congestion-in-score bends curves** — minimal Path II POC. **Not** bin-for-bin replication of empirical `poolq_loo`.

Say: "I'm not claiming we've estimated full **B** and **D** as functions of pool quality. Alex's score isn't claiming to be the whole **L_net** either. It's the **minimal generative piece** that makes congestion matter for **who gets selected**."

A.4 Thirty-second script (practice this)

> *"We use **one ontology, two readouts**.*
> *****L_net = B − D**** is my reduced form: strong peers can **help** through development and visibility and **hurt** through congestion and substitutability — that's the inverted-U story.*
> ***Alex's score** is narrower: **ability minus viable-peer congestion** in the **selection rule**. That operationalizes the **constraint leg** — who stands out when everyone around you is also viable — not the full benefit-minus-constraint decomposition.*
> *Empirically we see the curve on **leave-one-out pool quality**; the simulation shows congestion in the score can produce non-monotone curves on a **different axis**, with the limitation stated explicitly. Same model family, different rung."*

A.5 What not to say (Alex will push back)

Avoid	Say instead
"Alex's equation **is** my model."	"Alex's score **nests inside** my frame as the D-leg in selection."
"We estimated **B – D**."	"We **decompose conceptually**; v1 has quality vs. congestion **columns**, not full **B(Q)–D(Q)**."
"The sim reproduces `poolq_loo`."	"The sim shows congestion **in the score** matters; empirical U is on **LOO quality**."
"**S_i** includes development benefits."	"**B** shows up in theory and in proxies like minutes; **not in** S_i****."

A.6 If he asks: "So are they the same or different?"

Short answer:

> *"**Same project, different job.***
> ***L_net** is the **environment story** — why elite pools can help on average but hurt at the top.*
> ***S_i** is the **selection story** — how congestion enters **ranking** when slots are scarce.*
> *The score is the **minimal generative implementation** of the constraint side, sitting under the broader **B – D** frame — not a competing model."*

Path II in one breath: not two models fighting — **one ladder** (phenomenon → minimal score POC → quality-vs-congestion measurements → predictions).

A.7 Quick reference — S vs L vs L_net (not the same symbol)

Symbol	Level	Plain English	In data / code
****S_i****	**Individual**	Ranking score: ability minus congestion penalty	**Latent** in generative sim (`538D` CELL 10); not observed in Army/MBB/tenure panels
****L_C** / λ·**L_C****	**Individual-in-pool**	Congestion channel — **D leg inside S**	`crowding_smooth` (empirical D proxy)
`poolq_loo` **/** L_Q****	**Individual-in-pool**	LOO **teammate quality** — **mixes B and D**	Empirical **x-axis** (Rung 1); often **decreasing** on sim Plot B when `SHOW_PLOT_B_TEAM_MEAN=False`
Pool mean / `team_mean`	**Pool**	Whole-roster average ability	Generative Plot B **x-axis** when `SHOW_PLOT_B_TEAM_MEAN=True` (539-style)
****L_net = B – D****	**Theory**	Why environment helps **and** hurts	Conceptual spine — **not** a column you regress directly in v1

Mnemonic: S = who (rank). **L = where** (pool context on the plot). ****L_net** = why** (help minus hurt).

Critical: `poolq_loo` **is not** ≈ **λ**·L_C. Congestion is one ingredient in selection; pool quality is a **mixed** environment readout.

A.8 Bridging the two plots (same Y, different X — how the formulas connect)

Your question: Alex side feels like score → draft; your side is pool quality → draft. How do we link them?

What Alex's equation is for — simulation vs. explanation (read this twice)

Your insight is correct. Alex's score is primarily a **simulation rule**, not a claim that this one equation **fully explains** your empirical inverted-U on `poolq_loo`.

Layer	Object	Job in v1	What we claim
Empirical (Rung 1)	Inverted-U on `poolq_loo`	**The phenomenon** — established Army + MBB; preliminary tenure	Pattern is **real** and **replicated** (honest tenure caveat)
Theory (spine)	`L_net = B − D`	**Why** help and hurt can coexist in the same peer environment	Conceptual decomposition — **not** separately estimated as B(Q)–D(Q) in v1
Generative (Rung 2)	`S = A − λ·L_C` → top K	**Minimal POC engine** — run a disciplined simulation	Congestion **in the score** matters; **talent-only fails** (λ = 0)

What the sim proves (and does not):

Alex sim does	**Alex sim** does not
Show that **ability-only selection** cannot produce the generative story you need	Claim to **reproduce** your empirical curve on `poolq_loo` bin-for-bin
Show that adding **congestion to the score** can **bend** aggregate selection curves (on **pool mean** in v1 POC)	Replace `L_net` as the explanation for why elite pools help on average but hurt at the top
Operationalize the **D-leg inside selection** — who stands out when peers are viable substitutes	Put the **B-leg** (development, visibility, minutes) **inside** the score equation
Earn its place as a **Wang-style minimal mechanism** supplement	Serve as the **only** model of the empirical inverted-U

One sentence: Alex's equation **runs** the generative world; `L_net` **frames** the empirical world. The sim asks: *"Is congestion in selection plausibly necessary?"* — not *"Does this score equal your observed U on LOO quality?"*

Path II honesty (say aloud): "We do **not** claim the simulation **is** the inverted-U. We claim the simulation shows a **minimal ingredient** — congestion in ranking — without which the generative story fails, while the **empirical** inverted-U lives on a **different conditioning axis** (`poolq_loo` vs pool mean), stated explicitly."

Both plots share one **Y**:

	Y-axis
Empirical (you)	P(drafted) or mean draft rate in bin
Generative (Alex POC)	P(selected) or mean selection rate in bin

Same object: **advancement probability**.

Different conditioning (X) — and a common confusion

Plot	Typical X-axis	What X means
Your empirical readout	`poolq_loo` (LOO pool quality)	Observable local environment — **B and D still bundled**
Alex generative readout (Plot B)	`team_mean` (pool mean) — Path II default	Pool-level ability after soft assignment; **not** individual **S_i** on the axis
Alex (alternate Plot B)	`poolq_loo` **/** L_Q****	LOO quality axis — closer to yours, but sim often **does not** reproduce empirical U with same knobs

Important correction: The main generative **figure** is usually **not** "score vs draft probability." ****S_i**** works **inside** the sim (rank everyone, take top **K**). The **published** generative plot is **"selection rate vs pool feature"** — same *kind* of figure as yours (environment on X, outcome on Y), but a **different pool statistic** on X (pool mean vs LOO quality).

At the **individual** level (behind the plot):

```
Higher S_i  →  higher chance i is in top-K  →  higher P(Y_i = 1)
```

At the **aggregated** level (what you actually draw):

```
Bin by pool environment L  →  plot mean P(Y) in each bin
```

The bridge formula (conceptual)

Think in two steps:

Step 1 — Micro (Alex selection rule):

```
S_i = A_i - λ L_C,i => Y_i = 1\{S_i in top K (or top )\}
```

Step 2 — Macro (your empirical stylized fact):

```
E[Y | L_Q] = mean draft rate in pools with LOO quality L_Q
```

Link:

```
E[Y | L_Q] ≈ E[1\{S top K\} | L_Q]
```

The **left** is your curve (phenomenon). The **right** is what the sim **aggregates** after running the score rule inside each pool. ****L_net = B − D**** explains **why** the left-hand curve can rise then fall: mid-quality pools net-help; top pools net-hurt once **D** dominates **B**. **λ**·L_C**** in **S** is only the **congestion channel** in the ranking step — not the full mixed **L** on your x-axis.

Side-by-side (print this)

	Empirical (Rung 1)	Generative (Rung 2)	Theory (spine)
Question	How does draft rate vary with **peer pool quality**?	Can **congestion in the score** bend selection curves?	Why help **and** hurt?
X	`poolq_loo`	`team_mean` (v1 POC axis)	—
Y	P(drafted)	P(selected)	—
Mechanism in prose	**L_net(L_Q) = B − D**	**S = A − λ·L_C** → top **K**	B leg + D leg
What we claim	Inverted-U **replicated** (Army, MBB)	Congestion **matters**; talent-only fails	Decomposition + limitations

One paragraph to say aloud (plot bridge)

> *"Both figures plot **advancement probability on Y**. Mine puts **leave-one-out pool quality** on X — that's the **empirical inverted-U** we're trying to explain. Alex's score doesn't go on the x-axis. It **runs inside** the simulation: rank everyone with **S**, take top K, then plot **selection rate against pool mean**. So at the micro level **S** governs who wins slots; at the macro level **L** on my x-axis is the pool environment I condition on.*

> *Here's the key distinction: `L_net = B − D` is my **why** — benefit and constraint in the environment. **Alex's score** is a **minimal simulation rule** — ability minus congestion in **ranking**. It shows congestion **matters** for selection (talent-only fails). It does **not** claim to be the full explanation of my empirical U on LOO quality, and it doesn't put the B-leg inside the score. The sim is a **proof-of-concept engine**; the inverted-U on `poolq_loo` is the **phenomenon**; `L_net` is the **conceptual bridge** between them."*

What would make them line up more (deferred — say honestly)

Full alignment would require the sim to reproduce `poolq_loo` bin-for-bin (LOO generative match — **parked**). v1 instead states the **axis mismatch** explicitly: same mechanism family, **different conditioning object** on X.

A.9 Thought experiment — same axis (`poolq_loo`), both show inverted-U

Bracket axis mismatch for a minute. Suppose Army is re-run on LOO pool quality and the inverted-U holds. Suppose the generative sim is also read on `poolq_loo` (or LOO-quality bins) and — with congestion in the score — also bends down at the top. **How do we explain that?**

Step 0 — What you're looking at (same picture twice)

Both sides plot the **same kind of object**:

```
X = LOO pool quality (poolq_loo)
Y = P(advance) or mean advancement rate in bin
Shape = rise through mid tiers, fall in elite tier
```

Empirical = **observed** officers / players / faculty. Generative = **simulated** agents after running the score rule. Same axes, same qualitative shape.

Step 1 — What the empirical curve is saying

Plain English: As the quality of your **local peer pool** (excluding yourself) goes up, advancement first goes up, then goes down.

What `poolq_loo` captures: A **mixed** readout of the peer environment — better teammates/colleagues on average, but also denser clusters of strong substitutes. You see the **net** curve; you do not yet see B and D separately on that axis.

`L_net = B − D` reading of the shape:

Region of X	Story (conceptual)
Low → mid pool quality	Net environment **helps** — visibility, norms, development upside from stronger peers (**B** dominates net)
Mid → elite pool quality	Net environment **hurts at the margin** — too many similarly strong, substitutable peers competing for scarce slots (**D** dominates net)

The inverted-U is the **signature** that both forces exist in the same local comparison pool.

Step 2 — What Alex's score does inside the aligned sim

Micro rule (unchanged):

```
S_i = A_i − λ·L_C,i     (ability minus viable-peer congestion)
Rank by S_i → top K get Y_i = 1
```

Why congestion alone can bend the curve on poolq_loo:

- In **higher** LOO-quality pools, more agents are **viable substitutes** (high A, above threshold).
- Congestion **L_C** is high → **S** is compressed for many people — especially **near-threshold** types who are good but not dominant.
- Even with strong own ability, harder to land in top **K** when everyone around you is also viable.
- **Aggregate** selection rate in the top pool-quality bins can **fall** even though the pool is "better."

Talent-only null ($\lambda = 0$): Score ≈ ability only → selection tracks talent → curve tends **monotone** (better pool → more high-A people → no elite-tier dip). **Fails** to match inverted-U. That is the generative falsification.

With $\lambda > 0$: Congestion enters ranking → non-monotone **selection vs pool quality** becomes **plausible on the same X** as the data.

Step 3 — How sim and data connect when axes match

Layer	Question	Answer when aligned
Data	Does advancement bend on LOO pool quality?	**Yes** — stylized fact (Army, MBB; tenure preliminary)
Sim	Can congestion-in-score produce that bend on the **same** X?	**Yes** — minimal mechanism POC (if knobs reproduce shape)
Theory	Why would both be true in one framework?	`L_net` — same ontology; sim operationalizes **D in selection**; environment story explains rise **and** fall

Bridge (aligned case):

```
E[Y | poolq_loo]  ≈  E[1{S in top K} | poolq_loo]    (same X, same Y kind)
```

Left = empirical inverted-U. Right = what the sim **aggregates** after running **S** inside each pool-quality bin.

Step 4 — What Alex's equation explains vs what `L_net` still adds

Even with perfect axis alignment, the jobs **do not collapse**:

	Alex score `S`	`L_net = B − D`
Explains downturn?	**Yes, via D in ranking** — congestion at the top	**Yes, via net constraint** — D dominates B in elite pools
Explains upturn?	**Only indirectly** — better pools may have more dispersion; not a B term in **S**	**Explicitly** — B leg (development, visibility, upside)
What's on the x-axis?	Same `poolq_loo` — still **bundles** B and D	Decomposition **concept** — not two separate estimated curves in v1
Generative role	**Simulation engine** — shows D-in-selection is **necessary** for the shape	**Narrative spine** — why the bundled x-axis **should** bend

Crisp sentence: Aligned axes mean sim and data tell the **same qualitative story on the same plot**. Alex's score explains **how congestion in ranking can generate the downturn.** `L_net` explains **why the whole environment curve rises then falls** (help and hurt in one frame) — including the part of the rise that is **not** inside the score equation.

Step 5 — What you would still not claim (even if aligned)

- That you **estimated** separate B(Q) and D(Q) functions
- That **S** includes development benefits (B-leg) — it does not
- **Parametric identity** — same shape ≠ same coefficients / same bins
- **Causation** — associational language throughout
- That one basketball generative preset **proves** Army and tenure

v1 claim (aligned version): Cross-domain **phenomenon** on LOO pool quality + **minimal generative ingredient** (congestion in score necessary; talent-only fails) on the **same conditioning axis** + `L_net` as the conceptual decomposition + honest limitations on tenure and estimation.

Thirty-second script (aligned-axis version)

> *"Imagine both plots use leave-one-out pool quality on X and advancement rate on Y, and both show the inverted-U. The data curve says: elite peer pools stop helping at the top. The simulation runs Alex's score inside each pool — ability minus congestion, top K selected — and can reproduce that downturn when congestion is on. Talent-only can't. So the sim shows congestion in **selection** is a **minimal necessary ingredient** for the shape. My `L_net` frame still adds the **why** in two legs: peers can help through development and visibility and hurt through substitutability and finite slots — the score puts only the constraint leg into ranking, not the full benefit-minus-constraint story."*

Print: `./scripts/convert_multiple_md_to_pdf.sh 30`
Print (dense): `./scripts/convert_multiple_md_to_pdf.sh --narrow 30`